

Exam

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E-Business and Network Security EHN410

9 July 2021



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Toetsinligting:			
Test information:			
Maksimum punte:		Volpunte:	
<i>Maximum marks:</i>	170	<i>Full marks:</i>	170
Duur van vraestel:	180 minute	Oopboek / toeboek:	Oop
<i>Duration of paper:</i>	<i>180 minutes</i>	<i>Open / closed book:</i>	<i>Open</i>
Totale aantal bladsye (hierdie blad ingesluit): 6			
<i>Total number of pages (including this page): 6</i>			

BELANGRIK- IMPORTANT

1. The examination regulations of the University of Pretoria apply.
2. Python may be used to aid in the answering of questions.
3. Carefully read the instructions on the next page.
4. No form of communication between students is allowed.
5. Please follow the Telegram student-to-lecturer communication rules as given by the lecturer.

Dosent(e):	1. Prof HC Myburgh
Lecturer(s):	2. Mr AJ Naude (AL)

Instructions

- Use the supplied *EXAM.py* file to code your answers in.
- Rename this file to *yourstudentnumber.py* (no “u” in front) and upload only this file to AMS at the end of the test.
- Use only the variable names supplied in *EXAM.py* for your answers. Incorrectly named variables will not be marked.
- Use only the following (list) format to store array variables:
 - DECIMAL: $x = [1,2,3,4]$
 - BINARY: $x = '1010'$
 - HEX: $x = '734AD4F3'$
 - LETTERS: $x = 'EXAMPLE'$
- Use only the following list format to store matrix variables:
 - DECIMAL: $x = [[1,2,3],[4,5,6],[7,8,9]]$
 - BINARY: $x = ['1010', '0110']$
 - HEX: $x = ['734AD4F3', 'A3B4F6E1']$
 - LETTERS: $x = ['EXAMPLE1', 'EXAMPLE2']$
- Use capital letters when storing HEX values.
- If you do not know the answer or did not have time to answer the question, please initialise the variables using this format: $x = []$ (do not use ‘x’ as in this example – use the supplied variable name!)
- ONLY GIVE YOUR ANSWERS IN *EXAM.py* – NO ADDITIONAL CODE PLEASE!

Question 1 [50]

The Blowfish algorithm is a symmetric encryption algorithm that is very efficient in its implementation and can be used as a drop-in replacement for DES and AES. It has a 64-bit block size and a key length of 32 bits. It is a 16-round Feistel cipher and uses large key-dependent S-boxes. The encryption algorithm is shown in figure 1.

The operation of the round function (Fn) is shown in figure 2. It receives a 32-bit input which is split in four 8-bit blocks. Each S-box takes 8 bits as input (the MS bits go to S-box 1, the LS bits go to S-box 4) and produces 32 bits as output. For each S-box the 8-bit input is used to select a 32-bit sequence from 2^8 rows (index 0 to 255). Note this is a modification from the actual Blowfish algorithm for ease of implementation. The initial S-boxes are given in *Exam.py*.

During each round, P-values are XORed with the left 32-bits. The initial P-values are given in *Exam.py*.

The output of S-box 1 and S-box 2 are XORed, the result of which is XORed with the output of S-box 3, the result of which is XORed with the output of S-box 4. This is also a modification from the actual Blowfish algorithm for ease of implementation.

The key is used to transform the P-values and S-boxes before encryption/decryption. The key is XORed once with each 32-bit row of the P-values and the S-boxes to create the final P-values and S-boxes to be used for encryption/decryption.

Implement the Blowfish algorithm, described above, that uses **four** encryption rounds (as well as the final round) to perform encryption and decryption:

$$\mathbf{K_1} = 659F94E9_{16}$$

1. Encrypt the following message using the modified Blowfish algorithm:

$$\mathbf{P_{11}} = 8890830432106C2A_{16}$$

2. Decrypt the following ciphertext using the modified Blowfish algorithm:

$$\mathbf{C_{12}} = B28BCB76F804598E_{16}$$

GIVE ALL ANSWERS IN HEXADECIMAL FORMAT

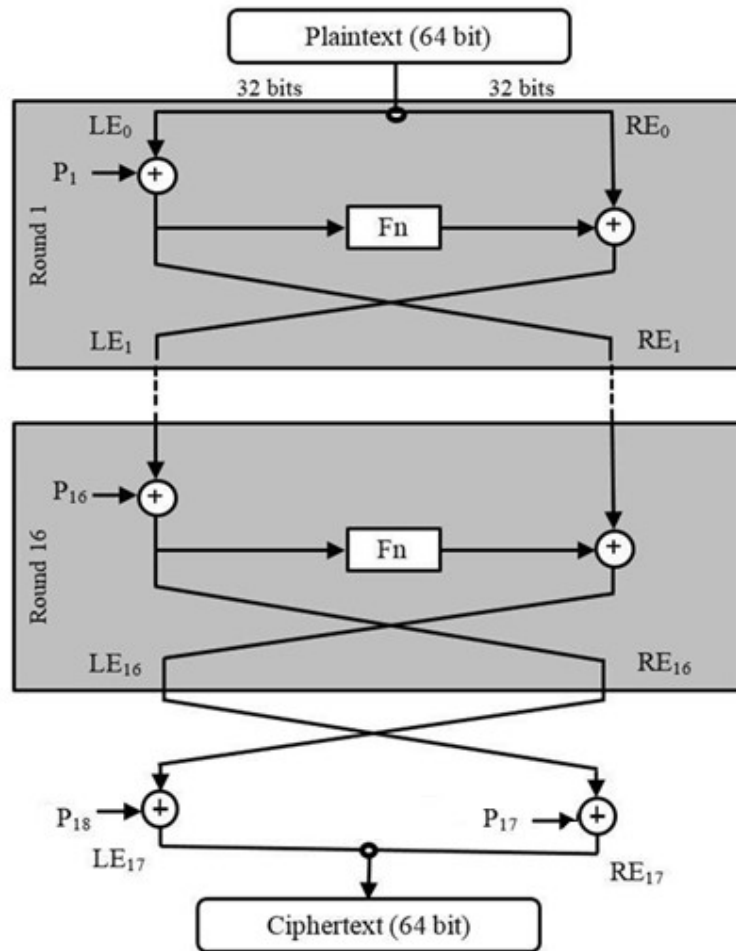


Figure 1 - Blowfish encryption algorithm

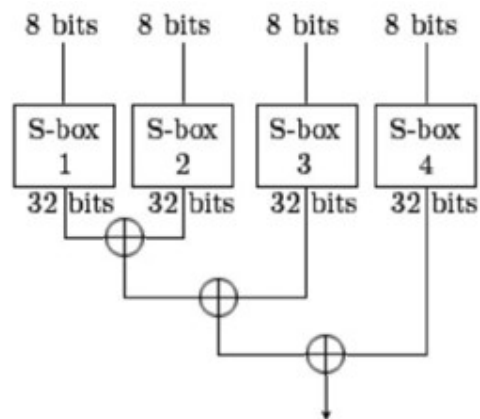


Figure 2 - Round function (Fn) operation

Question 2 [30]

RC4 is used to generate a random key stream in order to create a stream cipher.

2.1. Use the key below to generate an RC4 keystream to encrypt the following plaintext:

$$\mathbf{K}_{21} = 40636\text{BA}3\text{CF}45\text{D}7\text{BDD}32\text{E}_{16}$$

$$\mathbf{P}_{21} = \text{C}51\text{A}66120\text{A}10\text{CA}0908997\text{BD}293032\text{B}84\text{D}10\text{D}0841_{16}$$

2.2. Use the key below to generate an RC4 keystream decrypt the following ciphertext:

$$\mathbf{K}_{22} = \text{EF}7288\text{A}2\text{AA}702\text{CB}1\text{EC}5\text{F}_{16}$$

$$\mathbf{C}_{22} = 2120208\text{CEC}193808014\text{D}848011\text{E}888\text{A}2521100\text{D}8_{16}$$

GIVE ALL ANSWERS IN HEXADECIMAL FORMAT

Question 3 [60]

A message $\mathbf{P}_3$ is transmitted from Alice to Bob. The message is signed by Alice after which the HMAC is calculated and appended. The resulting message is encrypted using cipher block chaining, using the custom cipher on the last page (IV=00000000₁₆). The message received by Bob is given below.

$$\mathbf{M}_3 = \text{ECC}87\text{B}335\text{A}336485\text{D}19\text{CD}9\text{B}13\text{B}8\text{F}62548\text{E}943467_{16}$$

The following information is known:

- RSA: $\text{PRa}=77$, $\text{PUa}=5$, $n=119$. Sixteen 8-bit integers are transmitted.
- HMAC: $b=32$, $L=4$, $n=32$, $\text{ipad}=36_{16}$, $\text{opad}=5\text{C}_{16}$, $\mathbf{K} = \text{F}75\text{B}92\text{CA}_{16}$, IV=87FEB45A₁₆, simple hash function is used.
- Custom cipher: $\mathbf{K}_3=4\text{E}0\text{EFD}31_{16}$

Verify the authenticity of the message and calculate the original plaintext $\mathbf{P}_3$ sent by Alice.

Tip: If you are uncertain of your cipher block chaining decryption answer, you may use the alternative decrypted message $\mathbf{X}_{3alt}$ in Exam.py to continue with the question, for 75% of the remaining marks.

GIVE ALL ANSWERS IN HEXADECIMAL FORMAT

Question 4 [30]

Key wrapping is used for key distribution between Alice and Bob. For this question a shortened key wrapping algorithm is used.

- The data block size is 16 bits (instead of 64).
- The number of data blocks is $n=2$ (instead of 4).
- The number of wrapping stages is $s=3n$ (instead of $6n$).

The custom cipher below is used for encryption and decryption.

1. Perform key wrapping to distribute the key K_{41} using the key encryption key K_{E41} :

$$K_{41} = 634B0136_{16}$$

$$K_{E41} = B2890380_{16}$$

2. Perform key unwrapping on the wrapped key K_{42W} using the key encryption key K_{E42} :

$$K_{42W} = A41675F30E41_{16}$$

$$K_{E42} = A6C2CD38_{16}$$

GIVE ALL ANSWERS IN HEXADECIMAL FORMAT

Custom cipher

For this question a custom cipher is used. It takes 32 bits of plaintext as input together with a 32-bit key. Encryption works as follows:

1. Swap the left and right halves of the input.
2. Perform a circular one bit left shift.
3. XOR the result of 2 above with the key.

The plaintext and key below was used to produce the ciphertext below:

$$P = 902081C0_{16}, K = 4E0EFD31_{16}$$

$$C = 4D8FDD70_{16}$$